

AI Model Routing Policy

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TL;DR — If a licensed user's M365 Copilot seat can do the task, use the seat: it is already paid for. Everything else routes to the cheapest approved model that meets its data classification and capability needs, with Foundry as the default lane. Routing is how cost, compliance and resilience become enforceable rather than advisory.

1. Purpose and problem statement

This policy defines how Mace Consult selects and routes AI model workloads. Version 2 adds the business case absent from v1: why routing exists, what it protects, and why Microsoft Foundry is the default routing lane. It responds to three problems that grow with adoption:

Cost. Current model prices span roughly 25x between tiers on Azure OpenAI: GPT-5.4 Nano at £0.15/£0.93 per million input/output tokens versus the GPT-5.5 flagship at £3.70/£22.22 (GPT-5.4 Mini, the workhorse mid-tier, is £0.56/£3.33). Batch processing halves cost and prompt caching cuts repeated context by up to 90% — but only if workloads are deliberately routed. Billing is per invocation and per token: without routing, spend scales invisibly with success. The more AI is adopted, the worse un-routed spend becomes.

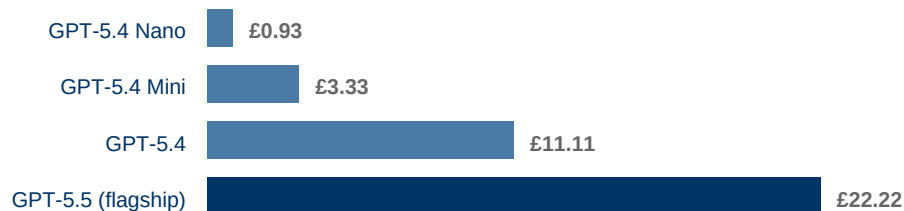


Figure 1 — The routing problem in one picture: output price, £ per million tokens, Azure OpenAI list rates (July 2026). A ~24x spread between the cheapest and most expensive tier — un-routed workloads default to the top bar.

Data classification. Not every model deployment is approved for every data class. Routing turns “client-confidential data goes only to approved endpoints” from guidance into an enforceable control, aligned with the G2 security and DLP gate in the tiered environments model.

Resilience and lifecycle. Models are deprecated on short notice cycles, rate-limited, and occasionally unavailable. Routing, together with the model-pinning register, is what prevents a model retirement from breaking a production application — a failure mode Mace has already experienced in the form of blocked AI change requests.

2. Use cases

Use case	Workload shape	Routing outcome
MaceStyle document validation	High-volume, repeated rule context, non-interactive	Small model + prompt caching + batch; frontier models add cost, not accuracy
Project chatbots (MaceyBot/ZoeBot)	Conversational, M365-grounded, internal data	Copilot Studio lane; escalate complex tasks to governed agents
Document summarisation at scale	Bulk, asynchronous, mixed data classes	Routed by data class; batch pricing; caching on shared context
Complex analysis / agentic workflows	Low-volume, high-capability, multi-step	Frontier model justified; per-invocation cost logged and capped

3. Business case

3.1 Worked example — MaceStyle validation (annualised)

Assumptions: 2,000 validations per month; ~15K input tokens per document (of which ~10K is reusable rule context) and ~2K output tokens. Substitute actual volumes from production telemetry; the ratio, not the absolute figure, is the point. Prices are Azure OpenAI pay-as-you-go list rates, July 2026, converted at £1 = \$1.35; Azure invoices in GBP will vary with the prevailing rate, and regional (data-zone) deployments carry a small uplift.

Routing approach	Cost per document	Annual cost
Un-routed: flagship (GPT-5.5, £3.70/£22.22 per MTok)	~£0.10	~£2,400
Routed: mid-tier (GPT-5.4 Mini, £0.56/£3.33)	~£0.015	~£360
Routed + rule context cached (90% off cached input)	~£0.010	~£240
Routed + cached + batch (50% off, overnight runs)	~£0.005	~£120

A single modest workload shows a ~20x spread between the default and the routed configuration. Applied across a portfolio of agents and per-invocation Copilot Credits, routing is the difference between AI spend that scales with value and spend that merely scales.

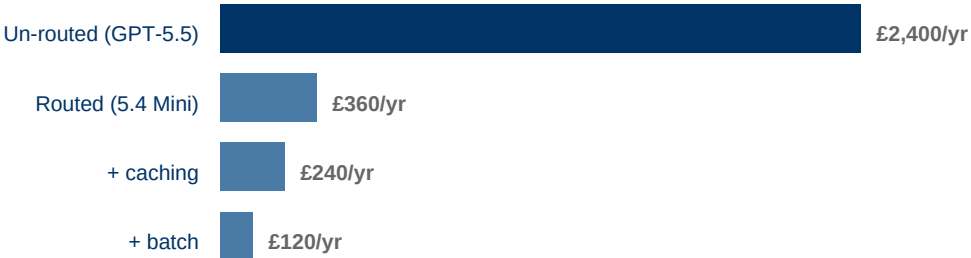


Figure 2 — MaceStyle validation, annual cost by routing approach (assumptions per 3.1).

3.2 Summary of benefits

Dimension	Without routing	With routing
Cost	Every workload defaults to the model someone picked; spend grows with adoption	Cheapest sufficient model per task; caching and batch by design; per-app attribution and caps
Compliance	Data-class rules are advisory	Approved endpoint per data class, enforced at the routing layer and evidenced at G2
Resilience	Model retirement or outage breaks apps reactively	Pinning register, notice-period handling, fallback lanes tested in staging

4. The routing framework

Four tiers route workloads by entitlement, data classification and interaction pattern. Tier 0 is new in v2, reflecting the enterprise-wide M365 Copilot rollout; Tiers 1–3 are unchanged from v1:

Tier	Route	Approved for	Notes
0	M365 Copilot (user seat)	Interactive tasks by licensed users on M365-resident data	Zero marginal cost — covered by the seat licence; always the first routing question
1	Azure OpenAI (GPT family)	Personal, client and regulated data	UK South / EU data zones; data residency enforced
2	Claude on Foundry (Zero Data Retention)	Internal, non-personal workloads	ZDR configuration; preferred for document intelligence and agentic tasks
3	Copilot Studio	Conversational assistants on M365 data	Permission-trimmed via Work IQ; maker-built agents subject to gate criteria

Seat-covered versus metered. The enterprise M365 Copilot rollout makes interactive colleague tasks a sunk cost: routing a licensed user's ad-hoc summarisation or drafting to a metered API pays twice for work the seat already covers. Conversely, the seat cannot serve programmatic pipelines, non-M365 data, high-volume batch work, or agents acting for unlicensed users and external processes — that is what Tiers 1–3 are for. Copilot Studio agents are metered per invocation (Copilot Credits) except where usage falls within a user's seat entitlement, so agent workloads are costed as Tier 3, not assumed free.

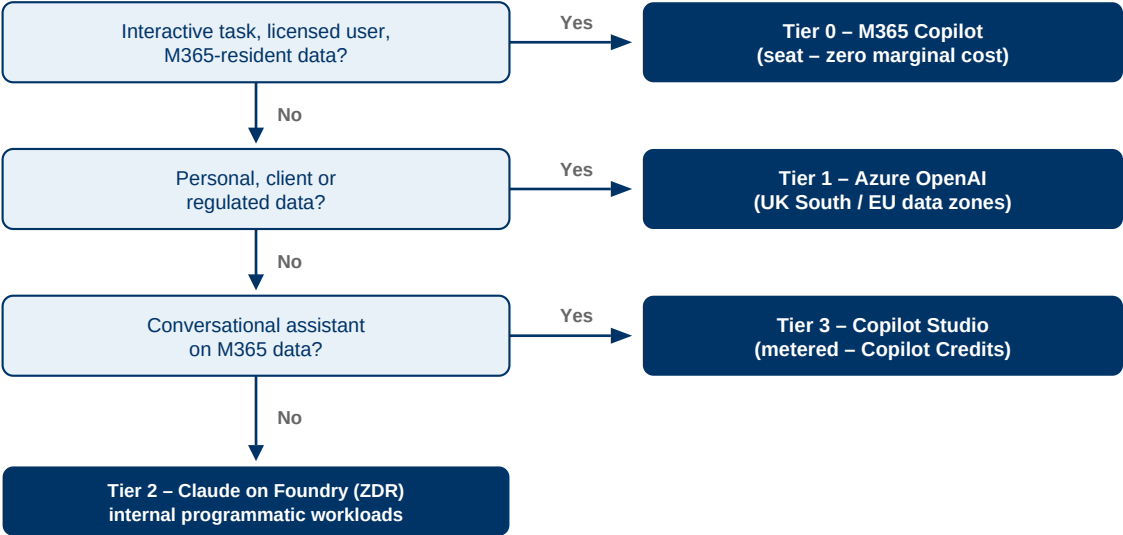


Figure 3 — Routing decision flow. Every new workload enters at the top; the first 'yes' determines the tier.

5. Routing within the Claude lane (Tier 2)

Routing applies within lanes as well as between them. Tier 2 workloads choose a Claude model on the same cost/capability logic as the GPT examples in section 3 — the price spread inside the Claude family is ~10x:

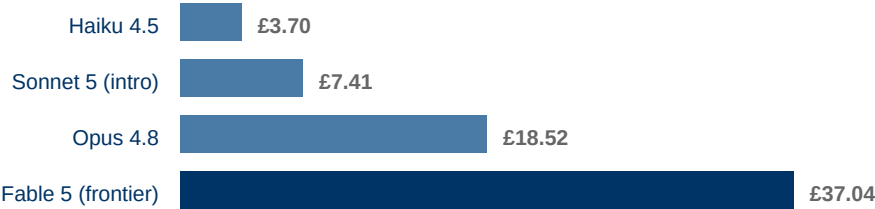


Figure 4 — Claude lane: output price, £ per million tokens, Anthropic list rates (July 2026), applied equivalently via Foundry. Sonnet 5 introductory pricing (£1.48/£7.41) runs to 31 August 2026; standard £2.22/£11.11 applies thereafter — a live example of why routing decisions are reviewed, not set once.

Workload scenario	Route to	Rationale
Document validation, extraction, classification (e.g. MaceStyle)	Haiku 4.5 + caching + batch	Highest-volume, lowest-complexity; frontier models add cost, not accuracy
Standard production workloads: summarisation, drafting, agent steps	Sonnet 5 (default)	Price/performance default for the lane; intro pricing until 31 Aug
Complex reasoning, multi-step agentic tasks, large-context analysis	Opus 4.8	Justified by task complexity; per-invocation cost logged and capped
Frontier-only tasks where failure cost exceeds token cost	Fable 5 – by exception	Requires named approval; recorded in the model-pinning register

The same levers apply as in the GPT lane: prompt caching (90% off cached input — MaceStyle's rule context is the canonical case), batch processing (50% off asynchronous work), and context trimming. Model deprecation notices and version pinning are handled through the model-pinning register per section 7.

6. Why Foundry is the default lane

Mace operates a Microsoft estate: Entra ID identity, Azure billing, Purview compliance, and an existing enterprise agreement. Foundry provides access to multiple model families — including Claude and GPT models — under the commercial, security and data-residency terms Mace already holds: one procurement relationship rather than several, and one control plane shared with the Microsoft Agent Framework workloads on the platform roadmap.

Foundry is the **default** lane, not an exclusive one. Direct Anthropic API access remains for capabilities Foundry does not carry, with Bedrock and Vertex retained as resilience fallbacks. Vendor concentration is a managed risk, reviewed alongside the model exposure matrix.

7. Governance and review

Routing decisions are recorded per application in the model-pinning register. New workloads declare their data classification and expected volume at G1; routing configuration is evidenced at G2; cost-per-invocation telemetry is a release criterion at G3. This policy is owned by the Digital AI Service, reviewed quarterly or on material model, pricing or platform changes.

v2.0 responds to review feedback (A. — v1 lacked use case and business case). v1 framework content is retained unchanged in section 4. Prices sourced from published Azure OpenAI pay-as-you-go rates, July 2026, converted at £1 = \$1.35; verify against the Azure pricing page and prevailing FX before external use.